

Impact Statement — Bridging the Tokenization Gap in African LLMs via SuperBPE

Challenge 3 submission · Team Manda · TRI AI Cohort 10 · Project 3

Field	Entry
Team	Law Adagbon — solo participant (no additional team members, no assigned mentor)
Project	Bridging the Tokenization Gap in African LLMs via SuperBPE
Related documents	Challenge 1 values-led problem statement; Challenge 2 Data Card (MATC v0.1)

What we are changing, and where the effect lands

A tokenizer is infrastructure. It sits beneath every model built on it, is chosen once, and is rarely revisited. Our project reduces how badly African-language text is fragmented before a model ever sees it, so the consequences of getting it right, or wrong, propagate into whatever is built on top and then persist.

Economic and access impact

Token counts set price, latency and how much context fits in a window. Because standard tokenizers shatter morphologically rich languages into meaningless fragments, the same sentence costs more in Yorùbá or Amharic than in English. That premium is charged per request, indefinitely, to the speakers least able to absorb it. Reducing fragmentation lowers the running cost of every downstream system for students, traders, teachers and the developers building for them, and because the artefact is small and freely reusable, anyone can adopt it.

Environmental impact

Training this tokenizer is cheap: CPU-only, hours, no pretraining run. Our own footprint is negligible. The leverage is downstream, where fewer tokens per request means proportionally less computation for the same conversation. We avoid overclaiming: efficiency gains often increase usage rather than reduce total consumption, so we lower the cost per interaction, not aggregate energy demand.

Cultural impact and its risks

A token vocabulary is a claim about what counts as a unit of a language. SuperBPE sharpens this, because superword tokens freeze frequent multi-word expressions into single units. Whichever varieties dominate our corpus become the ones the vocabulary treats as natural, which risks entrenching standard and prestige forms over regional ones. The evaluation setting compounds this: it flattens text to ASCII, collapsing Yorùbá words that differ only by tone and reducing Ge'ez and Ajami scripts to marker sequences. We mitigate by keeping diacritized and undiacritized text as separate strata, setting register quotas so conversational and code-switched writing is present, and releasing the vocabulary in readable form so speakers can see what it learned about their language.

Social and ethical risks

Three risks concern us most. **Lock-in:** because tokenizers are adopted and left alone, any bias we encode outlives our attention to it. **Widening internal gaps:** we can cover a few dozen languages out of more than a thousand, so success makes the included languages cheaper while leaving the rest relatively further behind. **Overclaiming:** cheaper tokenization could be marketed as African-language support by systems whose actual quality in those languages remains poor, letting an efficiency gain stand in for genuine capability.

Alignment and accountability

These decisions follow the values in our problem statement — equity, cultural respect, inclusion and accountability. We report results per language rather than as an average, because an average hides who lost; publish failures alongside gains; claim only the languages we cover; maintain a named route for a community or publisher to request removal of a source; and version our documentation so superseded claims stay visible.